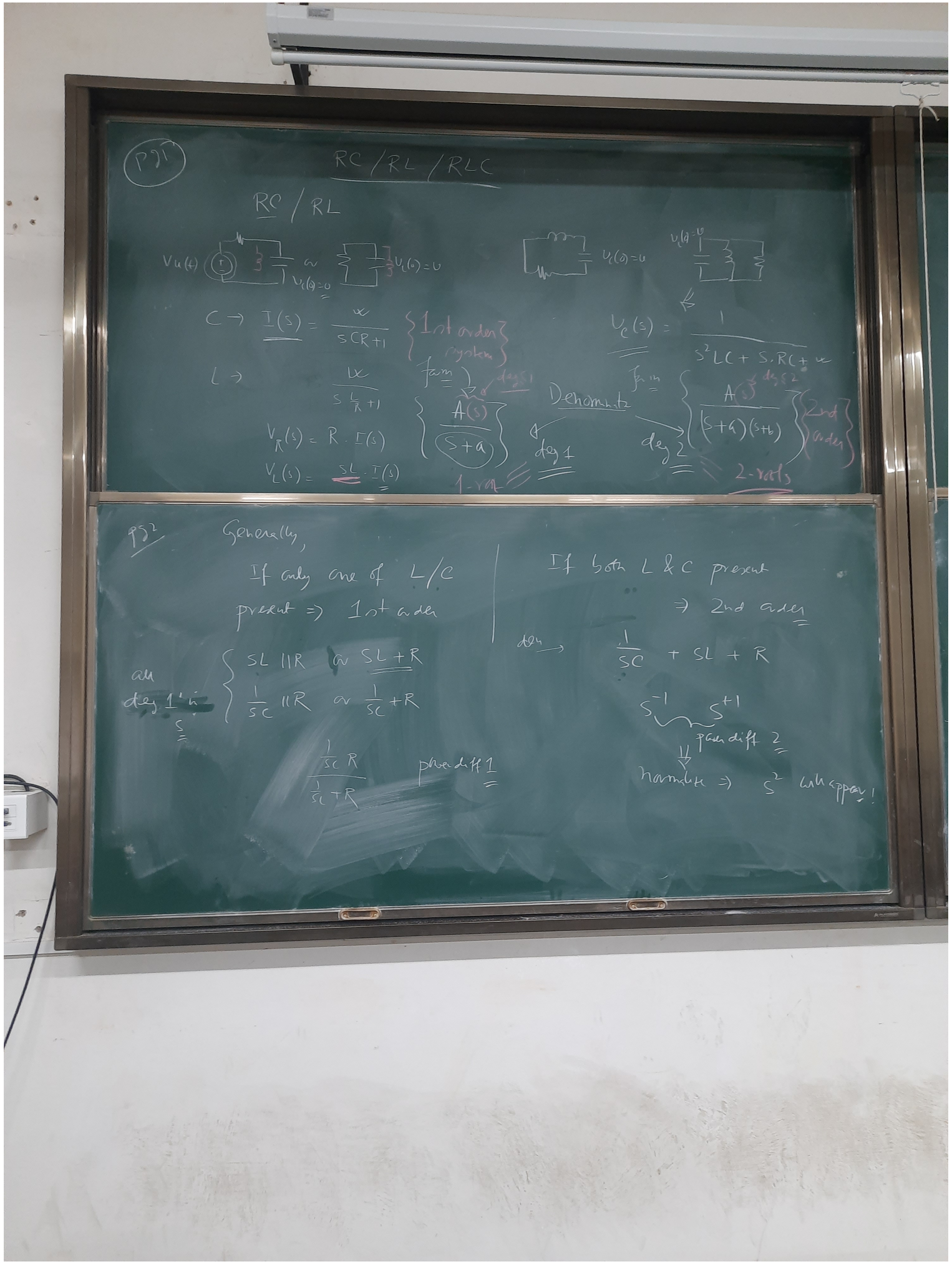


pg1

RC / RL / RLC

RC / RL

$C \rightarrow \underline{I(s)} = \frac{u}{sCR+1}$ {1st order system}

$L \rightarrow \frac{u}{s\frac{L}{R}+1}$

form → $\left\{ \frac{A(s)}{S+a} \right\}$ deg s 1

$V_R(s) = R \cdot I(s)$

$V_L(s) = SL \cdot I(s)$

1-root

Denominator

deg 1

deg 2

$\underline{\underline{V_c(s)}} = \frac{1}{s^2LC + S.RC + u}$

form $\left\{ \frac{A(s)}{(s+a)(s+b)} \right\}$ deg s 2 {2nd order}

2-roots

pg2

Generally,

If only one of L/C present ⇒ 1st order

all deg 1 in $\underline{\underline{s}}$ $\begin{cases} SL \parallel R & \text{or } \underline{\underline{SL+R}} \\ \frac{1}{sC} \parallel R & \text{or } \frac{1}{sC}+R \end{cases}$

$\frac{\frac{1}{sC}R}{\frac{1}{sC}+R}$ power diff $\underline{\underline{1}}$

If both L & C present

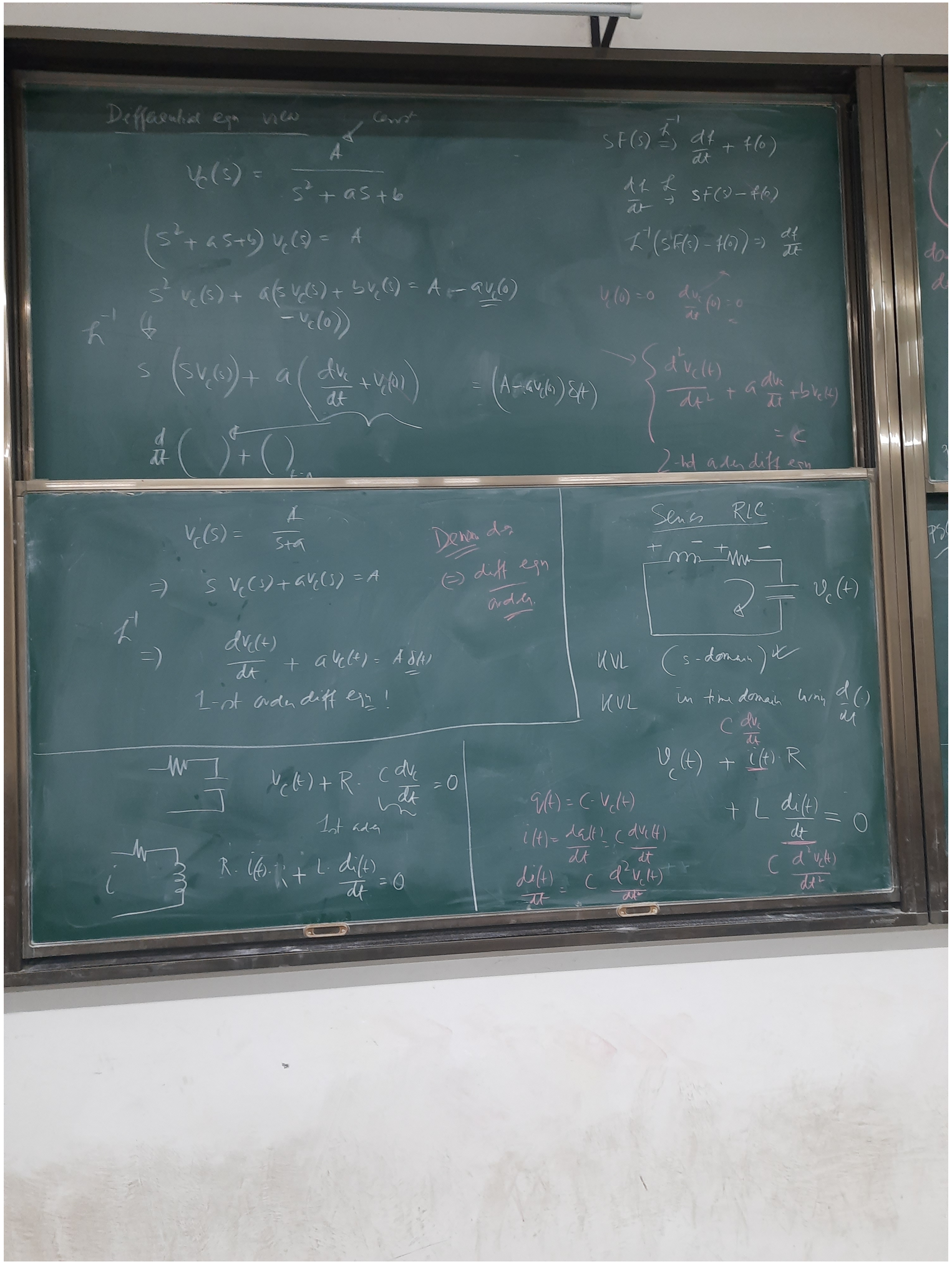
⇒ 2nd $\underline{\underline{\text{order}}}$

den → $\frac{1}{SC} + SL + R$

$\underbrace{S^{-1} \quad S^{+1}}$ power diff $\underline{\underline{2}}$

⇓

normalize ⇒ S^2 will appear!



Differential eqn view

const

$$V_c(s) = \frac{A}{s^2 + as + b}$$

$$(s^2 + as + b)\,V_c(s) = A$$

$$s^2 V_c(s) + a(sV_c(s) - v_c(0)) + bV_c(s) = A - a\underline{\underline{v_c(0)}}$$

$\mathcal{L}^{-1}$ $\Downarrow$

$$s\,(sV_c(s)) + a\underbrace{\left(\frac{dv_c}{dt} + v_c(0)\right)}_{} = (A - a v_c(0))\,\delta(t)$$

$$\frac{d}{dt}(\quad) + (\quad)_{t=0}$$

$$sF(s) \xrightarrow{\mathcal{L}^{-1}} \frac{df}{dt} + f(0)$$

$$\frac{df}{dt} \xrightarrow{\mathcal{L}} sF(s) - f(0)$$

$$\mathcal{L}^{-1}(sF(s) - f(0)) \Rightarrow \frac{df}{dt}$$

$$v_c(0) = 0 \qquad \frac{dv_c}{dt}(0) = 0$$

$$\rightarrow \left\{ \frac{d^2 v_c(t)}{dt^2} + a\frac{dv_c}{dt} + b v_c(t) = c \right.$$

2-nd order diff eqn

$$V_c(s) = \frac{A}{s+a}$$

$$\Rightarrow \quad sV_c(s) + aV_c(s) = A$$

$\mathcal{L}^{-1}$

$$\Rightarrow \quad \frac{dv_c(t)}{dt} + a v_c(t) = A\underline{\underline{\delta(t)}}$$

1-st order diff eqn!

Denom. deg

$\Leftrightarrow$ diff eqn order

$$v_c(t) + R\cdot \underbrace{C\frac{dv_c}{dt}}_{} = 0$$

1st order

i

$$R\cdot i(t) + L\cdot\frac{di(t)}{dt} = 0$$

Series RLC

+ − + −

$v_c(t)$

KVL (s-domain) ✓

KVL in time domain using $\frac{d}{dt}(\cdot)$

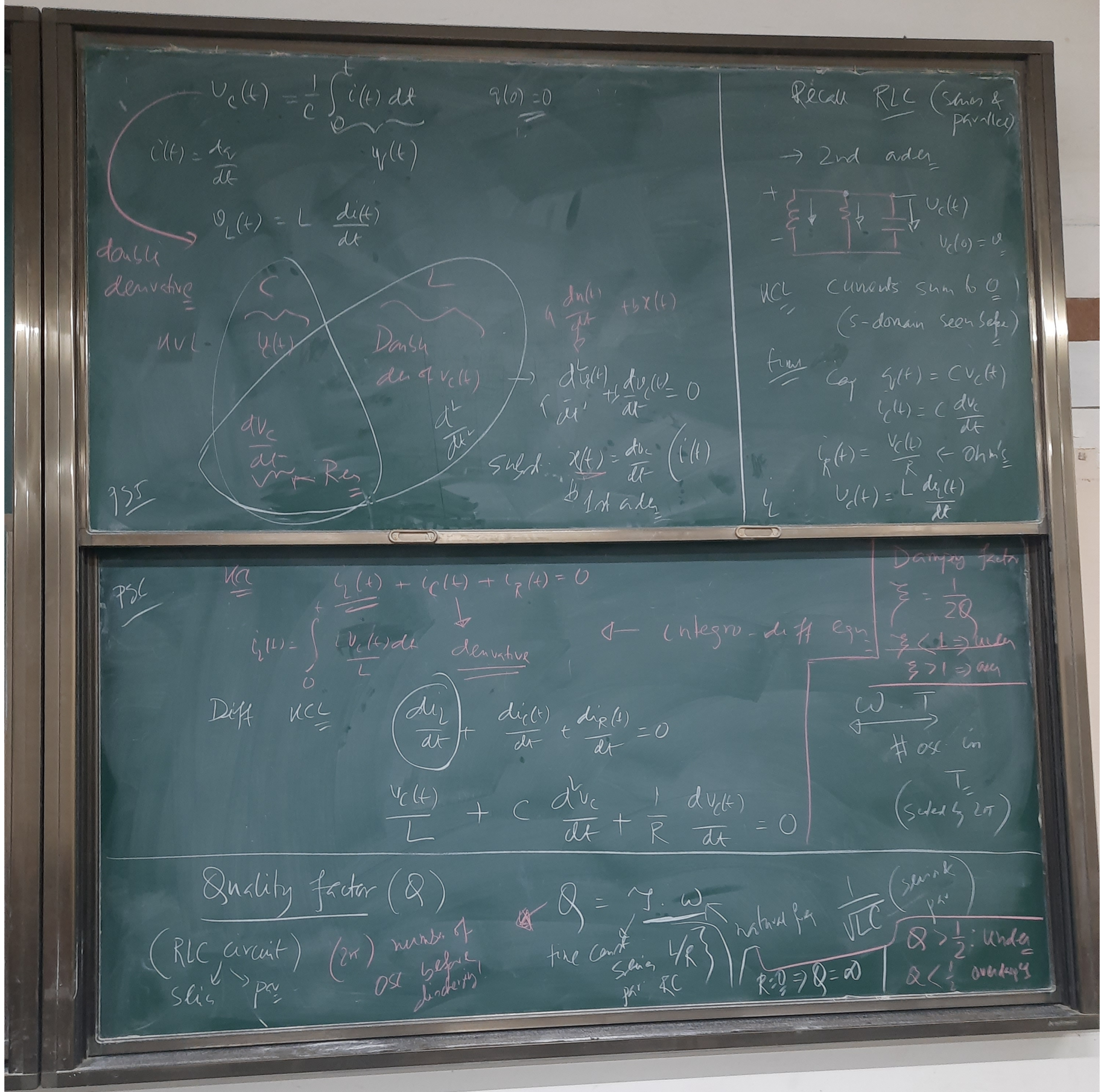
$$v_c(t) + \underline{\overset{C\frac{dv_c}{dt}}{i(t)}}\cdot R$$

$$+ L\underbrace{\frac{di(t)}{dt}}_{C\frac{d^2 v_c(t)}{dt^2}} = 0$$

$$q(t) = C\cdot v_c(t)$$

$$i(t) = \frac{dq(t)}{dt} = C\frac{dv_c(t)}{dt}$$

$$\frac{di(t)}{dt} = C\frac{d^2 v_c(t)}{dt^2}$$



$v_c(t) = \frac{1}{C}\underbrace{\int_0^t i(t)\,dt}_{q(t)}$

$q(0) = 0$

$i(t) = \frac{dq}{dt}$

$v_L(t) = L\frac{di(t)}{dt}$

double derivative

KVL

C

q(t)

L

Double der of $v_c(t)$

$\frac{dv_c}{dt}$

v_R Res

$\frac{d^2}{dt^2}$

$a\frac{dx(t)}{dt} + bx(t)$

$\frac{d^2 q(t)}{dt^2} + b\frac{dv_c(t)}{dt} = 0$

Subst: $x(t) = \frac{dv_c}{dt}$ ($i(t)$)

→ 1st order

PSS

Recall RLC (series & parallel)

→ 2nd order

$v_c(0) = v$

KCL currents sum to 0 (s-domain seen before)

Form: Cap $q(t) = Cv_c(t)$

$i_c(t) = C\frac{dv_c}{dt}$

$i_R(t) = \frac{v_c(t)}{R}$ ← Ohm's

i_L: $v_c(t) = L\frac{di_L(t)}{dt}$

PSC

KCL $i_L(t) + i_c(t) + i_R(t) = 0$

$i_L(t) = \int_0^t \frac{v_c(t)}{L}dt$

derivative

← integro-diff eqn

Diff KCL $\boxed{\frac{di_L}{dt}} + \frac{di_c(t)}{dt} + \frac{di_R(t)}{dt} = 0$

$$\frac{v_c(t)}{L} + C\frac{d^2 v_c}{dt^2} + \frac{1}{R}\frac{dv_c(t)}{dt} = 0$$

Damping factor

$\xi = \frac{1}{2Q}$

$\xi < 1 \Rightarrow$ under

$\xi > 1 \Rightarrow$ over

$\omega \cdot \tau$ ↔ # osc in τ (scaled by 2π)

Quality factor (Q)

(RLC circuit → series, par)

(2π) number of osc before dissipating!

$Q = \tau \cdot \omega$

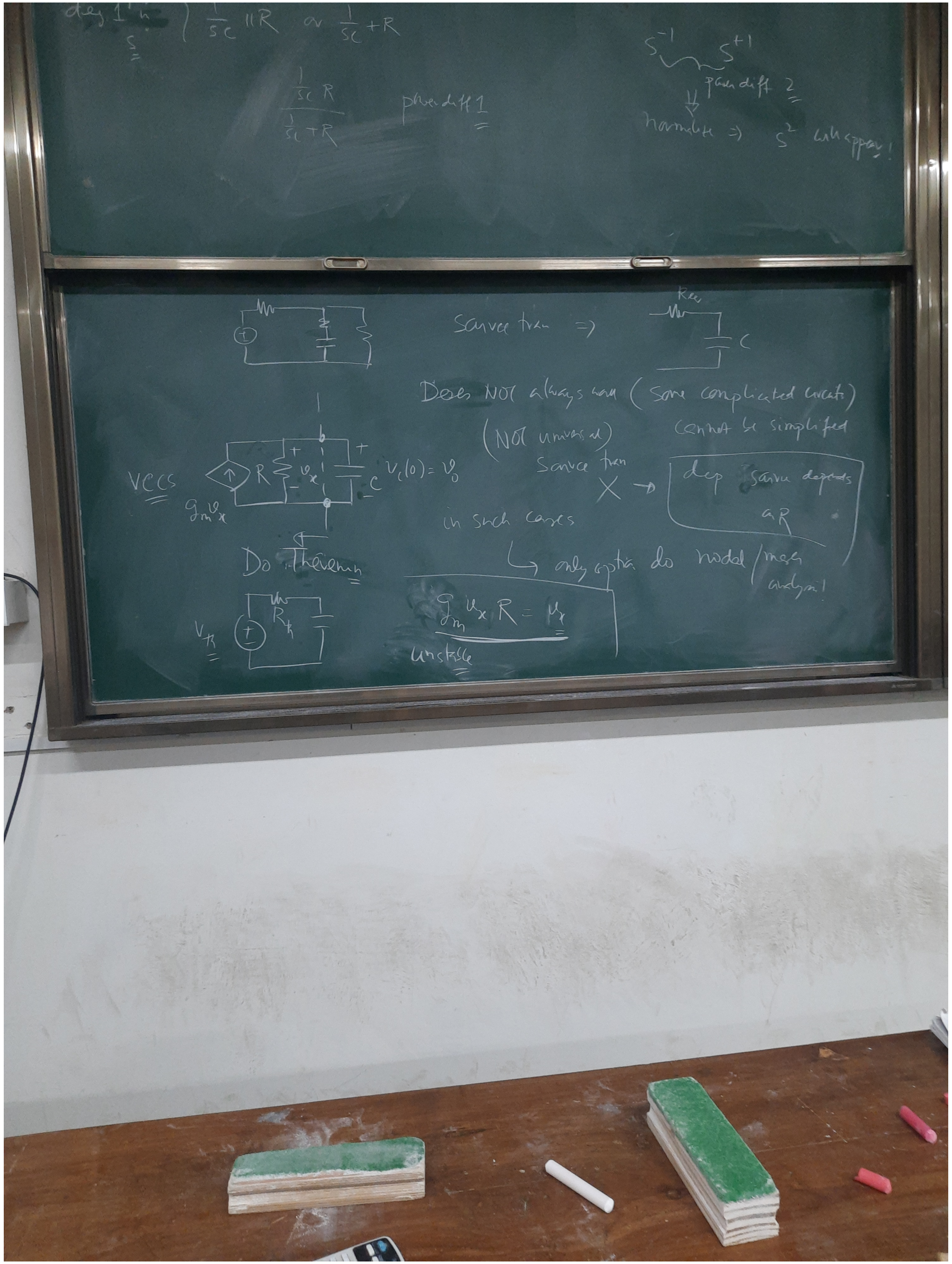
time const: series L/R, par RC

natural freq $\frac{1}{\sqrt{LC}}$ (series & par)

$R = 0 \Rightarrow Q = \infty$

$Q > \frac{1}{2}$: under

$Q < \frac{1}{2}$ overdamping



deg 1 in s
$\frac{1}{sC} \parallel R$ or $\frac{1}{sC}+R$
$\frac{\frac{1}{sC}R}{\frac{1}{sC}+R}$
power diff 1
s^{-1} s^{+1}
power diff 2
normalize ⇒ s^2 will appear!
Source tran ⇒
R_{eq}
C
VCCS
$g_m v_x$
R
v_x
C
$v_C(0)=v_0$
Do Thevenin
V_{th}
R_{th}

Does NOT always work (some complicated circuits) cannot be simplified

(NOT universal)

Source tran X → dep source depends on R

in such cases

↳ only option do nodal/mesh analysis!

$g_m v_x R = v_x$

unstable